

Sport to pave way for 3D

3D content delivered to television screens at home will probably begin with broadcasts of sport events. Many current events are already being recorded with 3D cameras

Forget HD, make way for 4k

The quality standards of video is spiralling upwards, and Blu-ray discs are keeping pace with increasing capacities. 1080p will soon make way for 4K resolutions



The Future of Immersive Entertainment

3D is not enough... How will you indulge your senses in the future?

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Truly immersive entertainment has always been somewhat the holy of the entertainment and electronics industry. Since the days of black and white cinema we have now come a long way in the ways in which we experience entertainment. We not only have stereo but surround. To experience visuals better we have high definition (HD). To actually feel some of the games we play there is force feedback. Now, we even have things like program sensitive ambient lighting and 3D movies. Shows a trend doesn't it? The aim here is to actually live the entertainment; be a part of the movie or the game in the truest sense. So what does the future hold for the sense-indulgent consumers of today? Let's take a peek...

3D technologies

3D is per se not the future because it is already here. There are several movies such as *Avatar* and *Alice in Wonderland* that are being released on 3D. In fact, at the time of going to print, we got an invite from Scrabble Entertainment to witness the IPL semifinal in 3D. Besides, you have games that support 3D through NVIDIA's 3D vision technology. But do you really like 3D in the form in which it's being dished out today? None of us want to wear those downright nerdy glasses every time we want to watch something in 3D. It's bearable if it's for a three hour event like *Avatar* but imagine wearing dorky glasses every time you sit to watch *Lost* or *Dexter*. Date killer, isn't it? Or imagine a gang of IPL fans jumping around in dork glasses every time Sachin shoots for the boundary. But don't despair, there is hope. Going by some of the products showcased at CES 2010 it's safe to say that the next big

revolution in entertainment is the 3D TV making its way to the home setting – and that too without those horrendous glasses.

3D is not however a new technology. Without sounding like a science lecture we can tell you it's been around since the turn of the last century (that's the late 1800s) in the form of anaglyph images. Anaglyph technology uses the principle of showing two images separately to your brain, which is the reason why we see in 3D in the first place. Each eye must see the image from a slightly different perspective, and the brain does the math to incorporate depth. So early on camera's were fitted with red and green (also cyan and blue) filters and different shots were taken from the perspective of the right and left eye separately. They were projected onto the same surface, so all the viewer had to do was wear glasses with red and green filters to fool the brain into thinking that two different perspectives were being seen – and voila – depth perception.

Hence you will find that while *Avatar* was great, so was *Chota Chetan* from yesteryear. Only on the bright side for *Avatar* there were fewer people having aneurysms and even fewer having headaches and dizzy spells. This is because of RealD – a new display technology system which uses circular polarization by which the flicker of traditional anaglyph is greatly reduced and viewing angles are increased. The company is currently working with many manufacturers to bring this technology to the home front. Still, this is how far 3D has matured. What about those dreams of 3D without facial attachments and ease of use?

Strangely, Intel showed off one of the most impressive demonstrations of glasses-less 3D technology on Alioscopy screens at CES this year. Reportedly, they'd set up a booth in which there were eight spots from where perfect depth was visible. These type of screens are being looked at by many

manufacturers around the world including NEC, Samsung and Phillips. Called auto-stereoscopic displays, they use an array of lenticular lenses to cast different images onto each eye thereby creating a perception of depth. Each lens covers 8 pixels and each sub-pixel is used for a different point of view. It therefore

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